

TABRIZ INTL, Iran, Islamic Republic of

WMO: 407060

Lat: 38.134N

Lon: 46.235E

Elev: 4459

StdP: 12.48

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.6	15.7	1.3	6.9	18.0	5.2	8.5	20.6	23.4	36.0	20.5	36.3	6.4	70	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.4	97.1	62.2	94.9	61.8	92.6	61.2	65.2	89.3	64.1	87.6	63.1	86.0	9.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.3	82.6	73.8	55.5	77.5	74.2	53.9	72.9	73.8	32.6	89.5	31.7	87.5	30.9	86.3	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.4	20.8	18.1	8.2	101.2	5.7	2.5	4.0	103.0	0.7	104.5	-2.5	105.9	-6.7	107.7		
			6.8	68.6	5.2	2.1	3.0	70.2	-0.1	71.4	-3.0	72.6	-6.8	74.1		
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	31.3	35.6	44.4	54.0	63.9	74.3	80.6	81.0	71.7	59.3	44.6	34.4
	DBStd	18.55	7.08	7.16	7.15	6.57	5.58	5.21	4.83	4.61	5.78	6.21	6.65	7.94
	HDD50	1888	580	403	200	32	0	0	0	0	0	8	181	484
	HDD65	4698	1045	823	638	333	88	4	0	0	13	193	612	949
	CDD50	4216	0	0	27	153	430	730	950	961	651	295	19	0
	CDD65	1551	0	0	0	4	54	283	485	496	214	15	0	0
	CDH74	16666	0	0	1	43	474	2825	5434	5664	2050	175	0	0
	CDH80	7683	0	0	0	6	72	1141	2764	2928	757	15	0	0
Wind	WSAvg	7.8	5.8	6.7	8.2	8.2	8.5	9.5	11.0	10.4	8.0	6.5	5.8	5.4
Precipitation	PrecAvg	9.7	0.8	0.6	1.1	1.9	1.6	0.4	0.3	0.2	0.2	0.8	1.1	0.8
	PrecMax	15.2	3.7	1.5	2.0	4.1	3.2	1.2	1.1	0.8	1.1	3.3	2.8	3.5
	PrecMin	6.1	0.0	0.0	0.0	0.6	0.3	0.0	0.0	0.0	0.0	0.1	0.0	0.0
	PrecStd	2.2	0.7	0.4	0.5	0.9	0.6	0.4	0.3	0.2	0.2	0.7	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	57.1	71.3	79.1	85.6	95.0	100.8	100.5	94.7	82.3	66.2	57.3
		MCWB	40.9	43.9	48.6	54.4	57.8	61.7	63.4	62.9	61.1	55.3	49.3	43.7
	2%	DB	46.8	53.2	64.7	73.5	82.0	91.7	98.2	97.2	90.0	78.8	62.4	51.8
		MCWB	38.5	41.6	46.8	51.8	56.5	60.3	63.1	61.8	59.5	54.2	46.8	41.8
	5%	DB	44.5	49.8	60.6	69.7	79.0	89.5	95.0	95.2	87.5	75.3	58.9	48.2
		MCWB	37.6	40.0	45.1	51.0	55.8	59.7	62.3	61.5	58.5	53.4	46.1	40.1
	10%	DB	41.3	46.5	56.9	66.2	76.6	86.3	91.8	93.0	84.4	71.8	55.5	44.8
		MCWB	35.7	38.0	43.5	49.5	55.1	58.7	61.8	61.1	57.6	52.3	45.0	38.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.5	45.7	50.7	56.4	61.3	64.5	67.3	66.4	63.6	57.8	52.3	45.7
		MCDB	48.7	54.9	64.8	72.3	78.8	88.4	93.6	90.4	86.5	74.5	59.5	52.9
	2%	WB	39.9	42.9	48.2	54.0	59.0	62.8	65.6	65.1	61.8	55.9	49.9	43.1
		MCDB	44.9	50.1	61.8	67.6	75.7	86.1	90.2	88.6	83.6	72.5	57.1	49.8
	5%	WB	38.0	40.8	46.2	52.7	57.4	61.5	64.5	63.9	60.3	54.5	47.8	41.0
		MCDB	43.0	47.0	58.3	65.5	74.1	83.8	88.4	87.2	81.6	70.5	55.5	46.2
	10%	WB	36.1	38.9	44.3	51.1	56.2	60.3	63.5	63.0	59.2	53.4	45.9	39.1
		MCDB	40.9	45.1	54.8	63.1	72.4	81.8	86.6	86.0	79.7	68.6	54.6	44.0
Mean Daily Temperature Range	5% DB	MDBR	14.5	16.4	18.2	19.0	20.7	22.2	21.7	22.4	22.8	20.7	17.3	14.6
		MCDBR	16.7	19.8	22.8	23.3	23.7	23.8	23.3	23.5	25.0	24.8	21.3	18.4
		MCWBR	10.3	11.2	11.1	9.8	8.5	8.2	7.9	7.5	8.9	10.4	11.5	11.4
	5% WB	MCDBR	15.1	17.2	20.6	19.9	21.8	22.6	22.4	22.5	23.5	21.8	17.5	16.1
		MCWBR	9.7	10.5	10.6	9.3	8.5	8.3	7.9	7.6	8.7	9.6	10.2	10.5
Clear-Sky Solar Irradiance	taub	0.311	0.346	0.401	0.471	0.459	0.434	0.470	0.436	0.387	0.390	0.332	0.307	
	taud	2.332	2.221	2.077	1.917	1.998	2.106	2.019	2.112	2.236	2.171	2.320	2.364	
	Ebn at Noon	273	276	270	257	263	269	258	264	272	256	262	266	
	Edh at Noon	29	37	48	59	56	50	54	48	40	39	30	26	
All-Sky Solar Radiation	RadAvg	772	1072	1320	1656	2058	2485	2420	2214	1805	1236	833	651	
	RadStd	59	115	115	151	105	125	82	77	93	92	98	85	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	+1.33	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.94	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page